

To estimate the second term, we use the fact that $\sum b_k$ converges, which implies $\sum_{k>K} \sum_{\ell=1}^{\infty} |a_{k\ell}| < \epsilon/2$ whenever $K > K_0$, for some K_0 . For the first term above, note that $\sum_{k \leq K} \sum_{\ell > L} |a_{k\ell}| \leq \sum_{k=1}^{\infty} \sum_{\ell > L} |a_{k\ell}|$. But the argument above guarantees that we can interchange these last two sums; also $\sum_{\ell=1}^{\infty} \sum_{k=1}^{\infty} |a_{k\ell}| < \infty$, so that for all $L > L_0$ we have $\sum_{\ell > L} \sum_{k=1}^{\infty} |a_{k\ell}| < \epsilon/2$. Taking $N > \max(L_0, K_0)$ completes the proof of (ii).

The proof of (iii) is a direct consequence of (ii). Indeed, given any rectangle

$$R(K, L) = \{(k, \ell) \in \mathbb{N} \times \mathbb{N} : 1 \leq k \leq K \text{ and } 1 \leq \ell \leq L\},$$

there exists M such that the image of $[1, M]$ under the map $m \mapsto (k(m), \ell(m))$ contains $R(K, L)$.

When U denotes any open set in $\mathbb{R}^2$ that contains the origin, we define for $R > 0$ its dilate $U(R) = \{y \in \mathbb{R}^2 : y = Rx \text{ for some } x \in U\}$, and we can apply (ii) to see that

$$A = \lim_{R \rightarrow \infty} \sum_{(k, \ell) \in U(R)} a_{k\ell}.$$

In other words, under condition (8) the double sum $\sum_{k\ell} a_{k\ell}$ can be evaluated by summing over discs, squares, rectangles, ellipses, etc.

Finally, we leave the reader with the instructive task of finding a sequence of complex numbers $\{a_{k\ell}\}$ such that

$$\sum_k \sum_{\ell} a_{k\ell} \neq \sum_{\ell} \sum_k a_{k\ell}.$$

[Hint: Consider $\{a_{k\ell}\}$ as the entries of an infinite matrix with 0 above the diagonal, -1 on the diagonal, and $a_{k\ell} = 2^{\ell-k}$ if $k > \ell$.]

3 Exercises

1. Suppose that $\{a_n\}_{n=1}^{\infty}$ is a sequence of real numbers such that the partial sums

$$A_n = a_1 + \cdots + a_n$$

are bounded. Prove that the Dirichlet series

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

converges for $\operatorname{Re}(s) > 0$ and defines a holomorphic function in this half-plane.

[Hint: Use summation by parts to compare the original (non-absolutely convergent) series to the (absolutely convergent) series $\sum A_n(n^{-s} - (n+1)^{-s})$. An estimate for the term in parentheses is provided by the mean value theorem. To prove that the series is analytic, show that the partial sums converge uniformly on every compact subset of the half-plane $\operatorname{Re}(s) > 0$.]

2. The following links the multiplication of Dirichlet series with the divisibility properties of their coefficients.

- (a) Show that if $\{a_m\}$ and $\{b_k\}$ are two bounded sequences of complex numbers, then

$$\left(\sum_{m=1}^{\infty} \frac{a_m}{m^s}\right) \left(\sum_{k=1}^{\infty} \frac{b_k}{k^s}\right) = \sum_{n=1}^{\infty} \frac{c_n}{n^s} \quad \text{where } c_n = \sum_{mk=n} a_m b_k.$$

The above series converge absolutely when $\operatorname{Re}(s) > 1$.

- (b) Prove as a consequence that one has

$$(\zeta(s))^2 = \sum_{n=1}^{\infty} \frac{d(n)}{n^s} \quad \text{and} \quad \zeta(s)\zeta(s-a) = \sum_{n=1}^{\infty} \frac{\sigma_a(n)}{n^s}$$

for $\operatorname{Re}(s) > 1$ and $\operatorname{Re}(s-a) > 1$, respectively. Here $d(n)$ equals the number of divisors of n , and $\sigma_a(n)$ is the sum of the a^{th} powers of divisors of n . In particular, one has $\sigma_0(n) = d(n)$.

3. In line with the previous exercise, we consider the Dirichlet series for $1/\zeta$.

- (a) Prove that for $\operatorname{Re}(s) > 1$

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s},$$

where $\mu(n)$ is the **Möbius function** defined by

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 \cdots p_k, \text{ and the } p_j \text{ are distinct primes,} \\ 0 & \text{otherwise.} \end{cases}$$

Note that $\mu(nm) = \mu(n)\mu(m)$ whenever n and m are relatively prime. [Hint: Use the Euler product formula for $\zeta(s)$.]

- (b) Show that

$$\sum_{k|n} \mu(k) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{otherwise.} \end{cases}$$

4. Suppose $\{a_n\}_{n=1}^{\infty}$ is a sequence of complex numbers such that $a_n = a_m$ if $n \equiv m \pmod q$ for some positive integer q . Define the **Dirichlet L -series** associated to $\{a_n\}$ by

$$L(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s} \quad \text{for } \operatorname{Re}(s) > 1.$$

Also, with $a_0 = a_q$, let

$$Q(x) = \sum_{m=0}^{q-1} a_{q-m} e^{mx}.$$

Show, as in Exercises 15 and 16 of the previous chapter, that

$$L(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{Q(x)x^{s-1}}{e^{qx} - 1} dx, \quad \text{for } \operatorname{Re}(s) > 1.$$

Prove as a result that $L(s)$ is continuable into the complex plane, with the only possible singularity a pole at $s = 1$. In fact, $L(s)$ is regular at $s = 1$ if and only if $\sum_{m=0}^{q-1} a_m = 0$. Note the connection with the Dirichlet $L(s, \chi)$ series, taken up in Book I, Chapter 8, and that as a consequence, $L(s, \chi)$ is regular at $s = 1$ if and only if χ is a non-trivial character.

5. Consider the following function

$$\tilde{\zeta}(s) = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s}.$$

- (a) Prove that the series defining $\tilde{\zeta}(s)$ converges for $\operatorname{Re}(s) > 0$ and defines a holomorphic function in that half-plane.
- (b) Show that for $s > 1$ one has $\tilde{\zeta}(s) = (1 - 2^{1-s})\zeta(s)$.
- (c) Conclude, since $\tilde{\zeta}$ is given as an alternating series, that ζ has no zeros on the segment $0 < \sigma < 1$. Extend this last assertion to $\sigma = 0$ by using the functional equation.

6. Show that for every $c > 0$

$$\lim_{N \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iN}^{c+iN} a^s \frac{ds}{s} = \begin{cases} 1 & \text{if } a > 1, \\ 1/2 & \text{if } a = 1, \\ 0 & \text{if } 0 \leq a < 1. \end{cases}$$

The integral is taken over the vertical segment from $c - iN$ to $c + iN$.

7. Show that the function

$$\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

is real when s is real, or when $\operatorname{Re}(s) = 1/2$.

8. The function ζ has infinitely many zeros in the critical strip. This can be seen as follows.

(a) Let

$$F(s) = \xi(1/2 + s), \quad \text{where} \quad \xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

Show that $F(s)$ is an even function of s , and as a result, there exists G so that $G(s^2) = F(s)$.

(b) Show that the function $(s-1)\zeta(s)$ is an entire function of growth order 1, that is

$$|(s-1)\zeta(s)| \leq A_\epsilon e^{a_\epsilon |s|^{1+\epsilon}}.$$

As a consequence $G(s)$ is of growth order $1/2$.

(c) Deduce from the above that ζ has infinitely many zeros in the critical strip.

[Hint: To prove (a) and (b) use the functional equation for $\zeta(s)$. For (c), use a result of Hadamard, which states that an entire function with fractional order has infinitely many zeros (Exercise 14 in Chapter 5).]

9. Refine the estimates in Proposition 2.7 in Chapter 6 and Proposition 1.6 to show that

$$(a) \quad |\zeta(1+it)| \leq A \log |t|,$$

$$(b) \quad |\zeta'(1+it)| \leq A(\log |t|)^2,$$

$$(c) \quad 1/|\zeta(1+it)| \leq A(\log |t|)^a,$$

when $|t| \geq 2$ (with $a = 7$).

10. In the theory of primes, a better approximation to $\pi(x)$ (instead of $x/\log x$) turns out to be $\text{Li}(x)$ defined by

$$\text{Li}(x) = \int_2^x \frac{dt}{\log t}.$$

(a) Prove that

$$\text{Li}(x) = \frac{x}{\log x} + O\left(\frac{x}{(\log x)^2}\right) \quad \text{as } x \rightarrow \infty,$$

and that as a consequence

$$\pi(x) \sim \text{Li}(x) \quad \text{as } x \rightarrow \infty.$$

[Hint: Integrate by parts in the definition of $\text{Li}(x)$ and observe that it suffices to prove

$$\int_2^x \frac{dt}{(\log t)^2} = O\left(\frac{x}{(\log x)^2}\right).$$

To see this, split the integral from 2 to $\sqrt{x}$ and from $\sqrt{x}$ to x .]

- (b) Refine the previous analysis by showing that for every integer $N > 0$ one has the following asymptotic expansion

$$\text{Li}(x) = \frac{x}{\log x} + \frac{x}{(\log x)^2} + 2\frac{x}{(\log x)^3} \cdots + (N-1)! \frac{x}{(\log x)^N} + O\left(\frac{x}{(\log x)^{N+1}}\right)$$

as $x \rightarrow \infty$.

11. Let

$$\varphi(x) = \sum_{p \leq x} \log p$$

where the sum is taken over all primes $\leq x$. Prove that the following are equivalent as $x \rightarrow \infty$:

- (i) $\varphi(x) \sim x$,
- (ii) $\pi(x) \sim x/\log x$,
- (iii) $\psi(x) \sim x$,
- (iv) $\psi_1(x) \sim x^2/2$.

12. If p_n denotes the n^{th} prime, the prime number theorem implies that $p_n \sim n \log n$ as $n \rightarrow \infty$.

- (a) Show that $\pi(x) \sim x/\log x$ implies that

$$\log \pi(x) + \log \log x \sim \log x.$$

- (b) As a consequence, prove that $\log \pi(x) \sim \log x$, and take $x = p_n$ to conclude the proof.

4 Problems

1. Let $F(s) = \sum_{n=1}^{\infty} a_n/n^s$, where $|a_n| \leq M$ for all n .

- (a) Then

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |F(\sigma + it)|^2 dt = \sum_{n=1}^{\infty} \frac{|a_n|^2}{n^{2\sigma}} \quad \text{if } \sigma > 1.$$

How is this reminiscent of the Parseval-Plancherel theorem? See e.g. Chapter 3 in Book I.